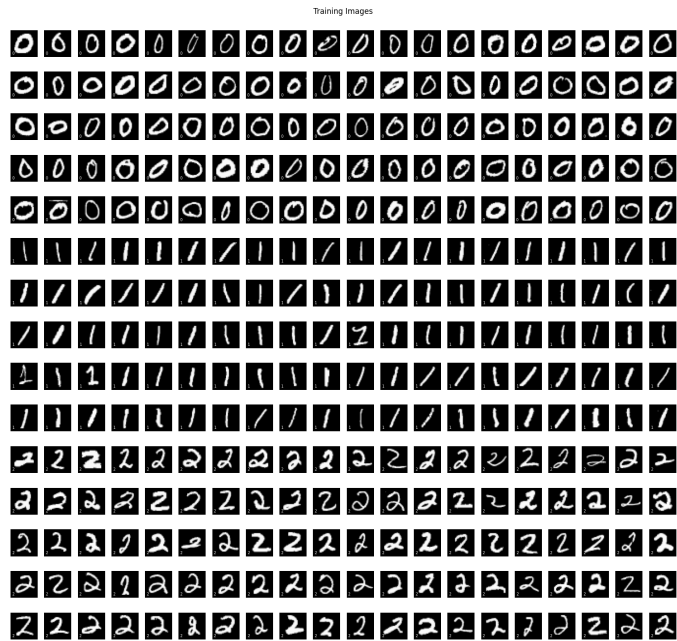


SML Assignment 1

Dataset

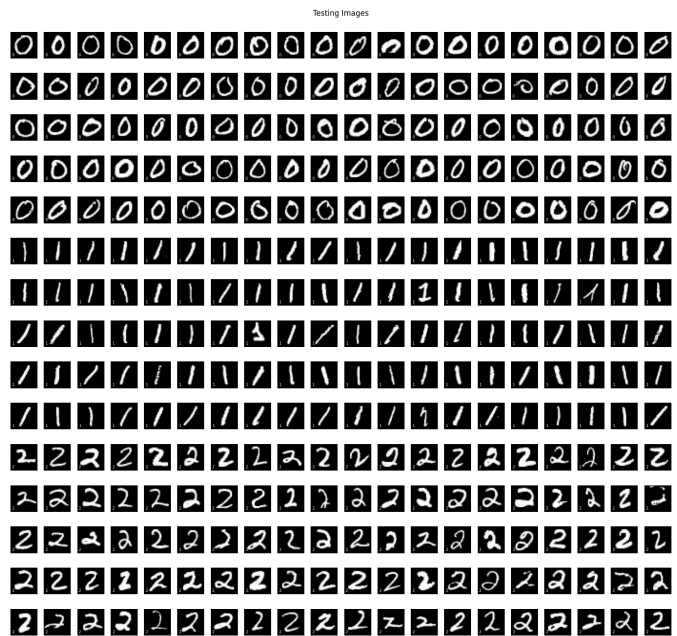
X_train: (300, 28, 28)

y_train: (300,)



X_test: (300, 28, 28)

y_test: (300,)



Accuracy Results

LDA accuracy: 0.9033

QDA accuracy: 0.9900

QDA significantly outperforms LDA on this dataset.

Model	Class	Precision	Recall	F1-score
LDA	0	0.922	0.94	0.931
LDA	1	0.866	0.97	0.915
LDA	2	0.930	0.80	0.860
QDA	0	0.990	1.00	0.995
QDA	1	0.990	0.99	0.990
QDA	2	0.990	0.98	0.985

Class-wise Precision, Recall and F1-scores further highlight this difference. While LDA shows reasonably balanced performance across classes, QDA achieves consistently high precision and recall for all three classes, indicating superior class separation.

To illustrate the decision mechanism, discriminant values were computed for a single test sample (belonging to class 0).

LDA discriminant values for one test sample:

Class 0: 73.7337

Class 1: -74.1704

Class 2: -8.3147

LDA assigns the test sample to Class 0, which has the highest discriminant value.

QDA discriminant values for one test sample:

Class 0: -3689.4010

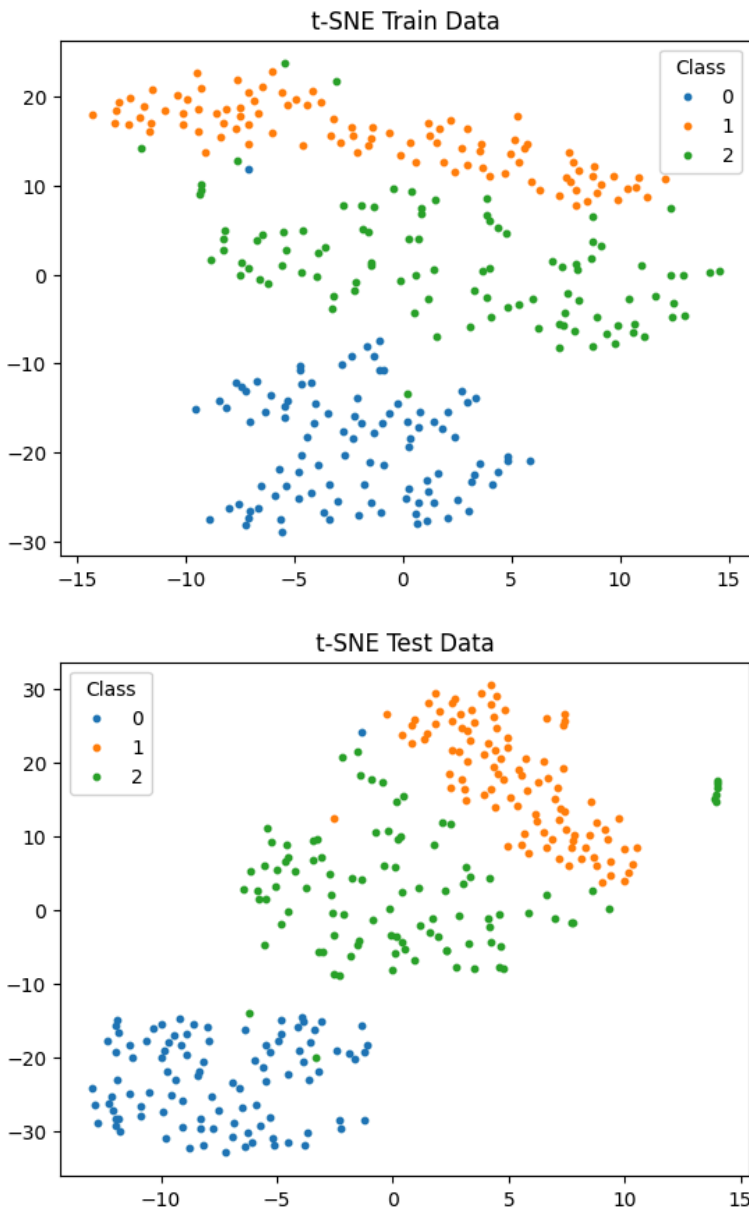
Class 1: -32419.6758

Class 2: -8443.8655

Similarly, QDA assigns the sample to Class 0, with a much larger separation between class scores due to the quadratic nature of the discriminant function.

QDA achieves higher accuracy than LDA because it models class-specific covariance matrices, allowing it to better capture the underlying distribution of each digit class. The strong performance of QDA suggests that the class covariances are sufficiently distinct and the applied regularization successfully mitigates numerical instability.

t-SNE Plots



t-SNE was used to project the high-dimensional feature vectors into 2-dimensions for visualization. The training data shows clear separation among the 3 digit classes, while the test data exhibits similar cluster structure with minor overlap. These visualizations are consistent with the strong classification performance, particularly for QDA.